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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

EE2003-1 – MICROCONTROLLER

(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Name the architecture and the instruction set for microcontroller?
A) Van- Neumann Architecture with CISC Instruction Set
B) Harvard Architecture with CISC Instruction Set
C) Van- Neumann Architecture with RISC Instruction Set
D) Harvard Architecture with RISC Instruction Set
2. External Access is used to permit _____
A) Memory interfacing
B) Peripherals
C) Power supply
D) ALE
3. CISC stands for
A) Complete Instruction Set Computer
B) Complex Instruction Set Computer
C) Compiler Instruction Set Computer
D) Complex Instruction Set Compiler
4. Internal RAM size of SST89x516RD2
A) 64k byte
B) 1k byte
C) 8k byte
D) 2k byte
5. TQFP stands for
A) Thin Quad Flip Package
B) Thick Quad Flat Package
C) Thin Quad Flat Package
D) Thick Quad Flip Package
6. In mode 1 operation, counter rolls over when it goes from _____ to _____
A) FFFFH to 0000H
B) 0000H to FFFFh
C) 00H to FFh
D) FFh to 00h
7. What is the operation for Timer in mode2?
A) Split Mode
B) 13-bit timer mode, 8-bit timer/counter THx and TLx as 5-bit prescaler
C) 8-bit auto reload mode, 8-bit auto reload timer/counter; THx holds a value which is to be reloaded into TLx each time it overflows16
D) 16 bit timer mode, 16-bit timer/counter THx and TLx are cascaded, no prescaler
8. To configure timer 1 in mode 1, the value to be loaded to TMOD register is
A) 0X01
B) 0X02
C) 0X20
D) 0X10
9. _____ timer register is bit addressable
A) TCON
B) TH0
C) TH1
D) TMOD
10. If transmission and reception are done simultaneously, then serial communication is called
A) Full Duplex
B) Half Duplex
C) simplex
D) Semi Duplex

11. Which of the SFRs is not used for serial communication
 - A) SCON
 - B) SBUF
 - C) PCON
 - D) SP
12. If timer 1 operates in mode 2, with reload value of FAH, SMOD=0, Fosc=11.0592MHz, then baud rate is
 - A) 9600
 - B) 4800
 - C) 1200
 - D) 19200
13. _____ register holds the serial data interrupt flag.
 - A) SCON
 - B) PCON
 - C) IE
 - D) IP
14. If the range of an 8 bit A/D converter is 0 to +5 V, then the resolution is
 - A) 12.5mV
 - B) 15.6mV
 - C) 0.625V
 - D) 19.53mV
15. In PWM technique, speed control of DC motor, _____ is true
 - A) Wider pulse, higher speed
 - B) Wider pulse, lower speed
 - C) Independent of pulse width
 - D) Higher speed at narrow pulse.
16. To drive a relay from microcontroller, _____ can be used
 - A) Transistor
 - B) resistor
 - C) Isolator
 - D) Direct Connection
17. Identify the temperature sensor in the list below
 - A) L293D
 - B) ULN2003
 - C) LM35
 - D) HC-05
18. What is ESP8266?
 - A) WIFI module
 - B) Temperature Sensor
 - C) Microcontroller
 - D) Bluetooth module
19. A driver circuit made up of PNP transistor can be turned on by applying
 - A) 0V
 - B) 5V
 - C) 3.3V
 - D) 2.5V
20. LED is always provided with series resistor
 - A) To increase the brightness of the LED
 - B) To limit the current flowing through the LED
 - C) To reduce power consumption
 - D) To improve the LED's colour

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Draw the internal and external memory structure of data memory of SST89X516RD2 microcontroller.	4	L*1	1	1
	b) Write a C program to add two 8 bit numbers A8h and 9Eh, display the sum in port 2 and carry in port 1 and Find the status of CY, AC and P of PSW after execution.	4	L2	1	1
	c) Explain the method and steps to program a timer in mode2 configuration.	8	L2	2	2
2.	a) Explain the internal structure of port 3 of microcontroller.	4	L1	1	1
	b) Write a c program to generate a 2kHz square wave on P2.0 with 50% duty cycle. Use Timer1 in Mode1. Assume crystal frequency of 12MHz.	8	L2	2	2
	c) Explain the bit field of Timer Mode Control Register (TMOD).	4	L1	1	1

3. a) Write a program to generate 2 square waves with one of 10kHz in P2.2 and another of 20kHz on P2.3. Use Timer in mode2. Assume $f = 12\text{MHz}$. Use interrupt.
- b) Explain the addressing modes with suitable examples.
- c) Draw the internal architecture of SST89X516RD2 microcontroller.

8	L2	2	2
4	L1	1	1
4	L1	1	1

Unit – II

4. a) Write a program to receive data 'A' serially with 9600 baud rate. Assume $f = 11.0592\text{MHz}$
- b) Write a program to display "DEPT OF EEE" on a 16x2 LCD. The data pins of LCD are connected to P1. The control pins Register select and Enable are connected to P3.6 and P3.7 respectively.

8	L2	3	3
			1,2,3
			,5,
8	L3	4	9,10

5. a) Write a program to display up counter 00 to 10 if active high switch SW1 connected to P3.4 is pressed and down counter 10 to 00 if active low switch SW2 connected to P3.3 is pressed in seven segment Display.
- b) Explain the bit fields of SCON and PCON registers used in serial communication.

			1,2,3
			,5,
8	L3	4	9,10
8	L2	3	3

6. a) Write a program to blink LED array connected to P2 at 1Hz 10 times. Use Timer1 Mode 1 configuration.

			1,2,3
			,5,
8	L4	4	9,10

- b) List the steps involved in programming the microcontroller to receive data serially and State the function of the RI flag in serial communication.

8	L1	3	3
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Unit – III

7. a) A chef is responsible for maintaining the oven temperature between 25°C and 30°C . The chef frequently checks the thermometer and either turns on the burner to increase the temperature or opens the door/activates a fan to cool it down. Design a suitable microcontroller-based temperature control system that performs similar operations automatically.

			1,2,3
			,5,
8	L4	5	9,10

- b) Design a microcontroller-based system that performs a automatic control of a DC motor to maintain its speed at 1000 RPM.

			1,2,3
			,5,
8	L4	5	9,10

8. a) A gardener wants to keep the soil in a farm adequately moist not too dry and not too wet. The gardener uses a moisture meter to check the soil and then either turns on the water pump or stops it to maintain about 60% moisture. Design a microcontroller-based system that performs this automatic watering task using suitable sensors and control logic.

			1,2,3
			,5,
8	L4	5	9,10

- b) Design a suitable microcontroller-based system that continuously monitors the water level in real time, displays it on a 16x2 LCD, and activates a buzzer when the water level reaches 10% and 90%.

			1,2,3
			,5,
8	L4	5	9,10

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
